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REGULAR CONTRIBUTION

Revisiting adversarial robustness in NIDS: a multi-dataset evaluation of test-time evasion attacks

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Abstract

Machine learning-based Network Intrusion Detection Systems (NIDS) remain highly vulnerable to test-time evasion (TTE) attacks, yet existing research relies heavily on legacy datasets and ignores the impact of evaluation scale. This study presents a systematic evaluation of TTE attacks using the Adversarial Robustness Toolbox (ART) across three modern, domain-diverse datasets: UNR-IDD, CICIOMT-2024, and UAV-IDS-2025. We evaluate a PyTorch-based fully connected neural network classifier and a 1D Convolutional Neural Network (CNN)-based classifier against white-box (AutoAttack, Auto-PGD, Carlini & Wagner) and black-box (HopSkipJump) threat models. Furthermore, we present a scalability analysis by evaluating performance across incremental test-set proportions from 10 to 100%. Results demonstrate severe susceptibility across models: despite achieving clean baseline accuracies up to 100%, Carlini& Wagner and HopSkipJump attacks reduce classification accuracy by nearly 80%. While 1D CNNs exhibit greater structural resilience than the PyTorch classifier, their adversarial robustness consistently deteriorates as the test set volume increases. These findings confirm that the evaluation scale and feature structure strongly influence adversarial performance, underscoring the urgent need for scalable evaluation standards and adaptive defenses in modern NIDS.

Keywords Network intrusion detection system (NIDS) · Test time evasion · Auto attack · Auto PGD attack · HopSkip jump attack · Carlini and Wagner attack · White-box attacks · Black-box attacks

1 Introduction

Test Time Evasion (TTE) attacks manipulate input samples at inference time to induce misclassification while preserving the underlying malicious functionality. In such attacks, adversaries introduce carefully crafted perturbations to input samples to mislead the model without altering the original malicious behavior.

TTE attacks are typically categorized into perfect-knowledge white-box scenarios, where the adversary has full access to the model's architecture and parameters, and limited-knowledge black-box or grey-box scenarios, where access to internal model information is partial or entirely restricted.

The adoption of machine learning (ML) and deep learning (DL) techniques in cybersecurity has significantly enhanced the capabilities of Network Intrusion Detection Systems (NIDS). These systems can process large volumes of network traffic and detect subtle patterns of malicious activity that traditional rule-based approaches may fail to capture. Biggio et al. [1] first demonstrated that gradient-based optimization can effectively craft adversarial samples at test time. Such vulnerabilities are particularly critical in cybersecurity, as minor feature manipulations can allow malicious traffic to evade detection without altering its operational intent.

TTE attacks are especially concerning because they target deployed models, making them highly practical in real-world settings. For instance, Liang et al. [30] conducted a test-time evasion attack against Google's Phishing Page Filter by reverse engineering the classifier using a technique known as classifier cracking. The authors recovered approximately 80% of the model's internal rules, enabling precise crafting of phishing webpages. By inserting benign features or removing

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malicious indicators, attackers achieved a 100% evasion rate in experimental evaluations. This case highlights the vulnerability of perfect-knowledge, client-side machine learning systems to TTE attacks.

A wide range of TTE algorithms has since been proposed, including Fast Gradient Sign Method (FGSM), Carlini and Wagner (C&W), Auto Attack, Projected Gradient Descent (PGD), and HopSkipJump (HSJ). To consolidate adversarial research, Nicolae et al. [2] introduced the Adversarial Robustness Toolbox (ART), an open-source library that provides standardized implementations of both white-box and black-box attacks. ART has facilitated systematic robustness evaluation across multiple domains. Recent studies, such as those by Merzouk et al. [5, 8], have examined evasion attacks on benchmark NIDS datasets including NSL-KDD, CIC-IDS, and UNSW-NB15.

Other works have extended adversarial strategies beyond ART; for example, Croce et al. [24] proposed a parameter-free extension of PGD, improving attack effectiveness against defended models, while Chen et al. [26] introduced targeted and untargeted variants of the HSJ attack, demonstrating its adaptability in black-box settings.

Although the literature on TTE continues to expand, several research gaps remain. First, many existing studies rely heavily on traditional benchmark datasets such as NSL-KDD, CIC-IDS, and UNSW-NB15. These datasets contain a limited number of attack categories, often dominated by conventional threats such as DoS, DDoS, spoofing, Mirai, and backdoor attacks, which may not fully represent modern and evolving attack landscapes.

Second, numerous prior works emphasize Attack Success Rate (ASR) as the primary evaluation metric. While ASR is informative, it does not provide a comprehensive comparison between clean and adversarial performance, thereby limiting insights into dataset quality and feature robustness.

Third, insufficient attention has been given to dataset characteristics such as feature distribution, correlation structure, and class complexity. Since identical models may behave differently across datasets, understanding these properties is essential for interpreting robustness outcomes. Finally, the scalability of adversarial evaluation, specifically how varying test set sizes influence adversarial performance, remains underexplored.

To address these limitations, this work presents a comprehensive evaluation of TTE attacks using the ART toolbox across diverse and contemporary datasets. We consider white-box attacks, including Auto-PGD, Auto Attack, and C & W, as well as the black-box HSJ attack. These attacks are applied to a convolutional neural network implemented in PyTorch, as well as the 1D convolutional neural network (CNN).

Our experimental evaluation spans datasets drawn from distinct operational domains. The UNR-IDD dataset incorporates fine-grained intrusion interpretation through network port-level characterization. The CICIOMT-2024 dataset focuses on medical Internet of Things (IoT) environments and includes communication protocols such as Wi-Fi, MQTT, and Bluetooth. The UAV-IDS 2025 dataset models intrusion scenarios in unmanned aerial vehicle (UAV) swarm networks and includes modern attack categories such as Blackhole, Flooding, Sybil, and Wormhole attacks, alongside normal traffic. Unlike traditional datasets centered primarily on DoS/DDoS-style attacks, these datasets reflect more contemporary and complex threat environments. This selection enables a deeper investigation of adversarial robustness across varied and evolving attack surfaces.

The primary contributions of this paper are summarized as follows:

- We evaluate TTE attacks on relatively recent and complex datasets like UNR-IDD, CICIOMT-2024, and UAV-IDS 2025 thereby extending adversarial robustness analysis beyond traditional benchmark datasets.
- We conduct a detailed comparative analysis of model behavior under both clean and adversarial conditions, enabling clearer interpretation of robustness degradation.
- Results are presented directly through confusion matrices and performance metrics, allowing transparent comparison between baseline and adversarial scenarios.
- While TTE was originally developed primarily for image-based domains, we extend its systematic evaluation to tabular network intrusion data involving modern attack categories.
- We investigate scalability by progressively increasing the test set size from 10 to 100% in 10% increments, thereby analyzing how evaluation scale influences adversarial robustness and model stability.

The remainder of this paper is organized as follows. Section II reviews related work on TTE attacks. Section III describes the proposed methodology and experimental setup. Section IV presents the empirical results and analysis. Section V discusses limitations and future research directions.

2 Literature review

Adversarial perturbations introduced during model testing can significantly mislead machine learning models, resulting in misclassifications. This behavior is commonly referred to as *Test-Time Evasion (TTE)* and has attracted substantial attention, particularly in machine learning-based Network Intrusion Detection Systems (NIDS). TTE can be conducted in both *white-box* and *black-box* settings, depending on the degree of access available to the target model.

In this study, we employ the Adversarial Robustness Toolbox (ART) to implement evasion attacks as proposed by Nicolae et al. [2]. ART offers a comprehensive framework for executing both evasion and poisoning attacks. Initially demonstrated on benchmark image datasets such as MNIST, the toolbox has subsequently been extended to additional image datasets including CIFAR and COCO. For example, Aditya et al. [13] evaluated CIFAR-10, MNIST, and ImageNet, achieving success rates of 99.5% and 80% on MNIST and CIFAR-10, respectively, using C&W attacks. Nazeri et al. [14] investigated object detection vulnerabilities in DETR models using MS-COCO and the KITTI dataset, applying C&W (L_1, L_2) and PGD attacks. Similarly, Juraev et al. [15] demonstrated that deeper architectures and more complex datasets such as ImageNet, CIFAR-10, and CIFAR-1000 require stronger perturbations to achieve successful evasion.

Peng et al. [16] introduced AutoAttack-based robustness improvements through flexible CNN scaling, convolutional stems with overlapping kernels, and delayed downsampling, enhancing performance across CIFAR and ImageNet. Chen et al. [17] applied Auto-PGD attacks in autonomous driving using the Comma2k19 dataset, integrating preprocessing, adversarial training, contrastive learning, and diffusion reconstruction, which resulted in short-range trajectory errors of up to 34.45 m. Xu et al. [18] proposed PatchZero, a defense mechanism that detects and removes adversarial patches by zeroing manipulated pixels. Their evaluation across image classification, object detection, and video classification tasks demonstrated strong robustness against patch-based attacks with minimal degradation in benign accuracy.

ART has also been widely used in computer vision research. Meyers et al. [20] tuned attack parameters to minimize perturbations while achieving misclassification in object detection using the DETR-ResNet50 model and CNN-based bone fracture classification.

Beyond image-based applications, ART has increasingly been adopted in network intrusion detection research. Awad et al. [3] evaluated DNN-based NIDS against C&W (L_2) attacks using CICIDS-2017 and CICIDS-2018 datasets. By incorporating adversarial training, Gaussian augmentation, and label smoothing, they mitigated attack effects and improved robustness across accuracy, precision, and recall metrics.

Muthalagu et al. [4] applied both C&W and Auto-PGD attacks on UNSW-NB15, CICIOMT-2024, and CICIOT-2023 datasets, expanding adversarial evaluation to more recent intrusion datasets.

Merzouk et al. [5] examined a broader spectrum of attacks, including FGSM, BIM, DeepFool, C&W (L_2, L_0, L_∞), and JSMA, across NSL-KDD, UNSW-NB15, and CIDDS-001 datasets. Their preprocessing pipeline, comprising one-hot encoding, Min–Max normalization, and binary label conversion, highlighted the susceptibility of NIDS to adversarial manipulation. In subsequent work, Merzouk et al. [8] evaluated reinforcement learning-based NIDS under FGSM and BIM perturbations, showing that even low-magnitude perturbations can substantially degrade detection performance.

Defense strategies such as Generative Data Augmentation (GDA) and Feature Squeezing (FS) have also been investigated. Roshan et al. [6] applied these methods against FGSM, PGD, and C&W attacks, demonstrating the effectiveness of preprocessing-based defenses in enhancing resilience.

Several studies have explored diverse classifiers under ART-based attack scenarios. Although ART provides a PyTorch classifier interface, Sharma et al. [7] evaluated Logistic Regression (LR), Multi-Layer Perceptron (MLP), Support Vector Machines (SVM), Decision Trees (DT), Random Forests (RF), XGBoost, and K-Nearest Neighbors (KNN) under white-box and black-box attacks, including Transfer, PGD-assisted Transfer, ZOO, and HSJ, on the NSL-KDD dataset.

To enhance robustness, Dinh et al. [9] proposed SAINT (a deep CNN with two convolutional and three dense layers) and attention-based TabNet architectures. These models were evaluated against evasion and poisoning attacks (FGSM, BIM, DeepFool, C&W) implemented via ART, with accuracy serving as the primary evaluation metric. Their preprocessing pipeline included data cleaning, missing value handling, and XGBoost-based feature selection.

Meyers et al. [10] further introduced massively parallel PGD attacks to assess KSVM models under full adversary knowledge assumptions. Their experiments incorporated adversarial retraining, variable batch sizes (1–1000), and adaptive perturbation step sizes to rigorously stress-test model robustness.

Kumar et al. [11] proposed constraint-based adversarial detection by leveraging feature interdependencies and domain constraints within the CICIDS-2017 dataset. Their approach successfully detected FGSM, BIM, PGD, DeepFool, and JSMA attacks, achieving near-perfect adversarial detection rates while maintaining minimal false positives.

Yang et al. [19] conducted PGD, transfer, and traffic morphing attacks on customized CICIDS-2017 datasets. They introduced three query strategies—Linear Query Matrix, Feature Space Traversal, and Dataset Migration—to construct high-fidelity substitute models via AutoML, successfully evading seven ML-based IDSs and one DL-based IDS with an average success rate exceeding 75%. Sun et al. [12] demonstrated 99.86% evasion success using HSJ attacks on the DAPT202 dataset.

Applications of ART have also extended to IoT and connected vehicle security. Aloraini et al. [21] and Gungor et al. [22] evaluated DNN, tree-based, and hybrid IDS models on IoT datasets such as MQTT, WUSTL-IIoT, and Edge-IIoT, as well as the HCRL car-hacking dataset.

Attacks including FGSM, BIM, PGD, DeepFool, JSMA, Elastic Net, Zeroth Order Optimization, and transfer-based methods were assessed under white-box, black-box, and gray-box scenarios, revealing substantial IDS vulnerabilities and emphasizing the need for layered defense strategies.

Despite the growing body of literature, several research gaps remain. Comparative analysis of dataset behavior across diverse adversarial attacks is limited. Moreover, while adversarial research is well established in image-based domains, systematic investigation within NIDS remains comparatively underdeveloped. Many existing studies rely on traditional datasets containing older attack categories, which may not reflect contemporary threat landscapes. Additionally, studies such as Gungor et al. [22] report drastic accuracy degradation from approximately 99% to as low as 4% under adversarial conditions, yet the underlying causes of such sharp declines remain insufficiently analyzed. The influence of dataset composition and data proportion on adversarial susceptibility warrants deeper statistical examination. The present work addresses these limitations by conducting a statistical analysis of adversarial attacks on newer intrusion datasets and examining dataset behavior under varying data partitioning strategies. Table 1

Table 5 provides a comprehensive summary of the previous works on Test-Time Evasion (TTE) attacks, including their experimental settings, datasets, and key results.

3 Methodology

The overall workflow of our study is divided into two primary components: (i) Attack Performance and (ii) Scalability Checking.

For attack performance evaluation, the models were initially trained using clean training data to establish baseline performance metrics. Subsequently, adversarial examples were generated from the testing dataset to conduct test-time evasion attacks. These adversarial samples were designed to mislead the trained models during inference. The resulting accuracy under adversarial conditions was then compared with the baseline performance to quantify the effectiveness of the attacks.

For scalability analysis, we systematically evaluated model robustness across varying proportions of the testing dataset. Specifically, subsets comprising 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, and 100% of the test data were selected iteratively. In this phase, a PyTorch classifier and a 1D CNN classifier were employed across all datasets to ensure consistency in experimental settings. The Adversarial Robustness Toolbox (ART) was utilized to implement evasion attacks in both the attack performance and scalability experiments. This approach enabled a comprehensive assessment of network intrusion detection system robustness under varying data scales.

Figure 1 illustrates the block diagram of the proposed scalability evaluation framework.

3.1 Datasets

ART is a recently adopted concept in network intrusion detection, and our work extends its evaluation across a diverse set of contemporary datasets, described in the following subsections.

3.1.1 CICIOMT-2024

We incorporate protocol-level IoT traffic by utilizing the CICIoMT-2024 dataset proposed by Dadkhah et al. [28], which contains several critical attack scenarios. This dataset contains stimulated and real data examples composed of 40 IoMT devices. The attacks contains, DDos, DoS, Recon, Spoofing and MQTT. These data were captured through WiFi/MQTT enabled IoMT devices and. A combination of malicious PC and a smartphone was used to capture malicious data and Bluetooth Low Energy (BLE) for benign data. This dataset provides real time duplication of packets and created for security and profiling which make this dataset suitable for our experiment.

3.1.2 UNR-IDD

The UNR-IDD dataset created by Das et al. [27] captures real-world intrusion samples. This dataset uses port statistics for fine-grained analysis of intrusion. Also, this dataset provides sufficient and equal representation of various anomaly types and adequate tail class representation. The labels this dataset present are Blackhole, Diversion, Normal, Overflow, Port Scan and TCP-SYN with 34 features. Logs were captured periodically on 5 s. This dataset was generated by Software-defined Network (SDN) simulation environment ensuring that the dataset is not dependent on any static topology and can be configured to reproduce the network activity of various topologies. These features makes this dataset robust in classifications and provides an appropriate data environment for our experiment.

3.1.3 UAV-IDS-2025

We have also used recent dataset UAV-IDS-2025 proposed by Zeng et al. [31], which is designed to evaluate intrusion detection systems in Unmanned Aerial Vehicle (UAV) swarm networks. It was generated through simulations utilizing the NS−3.24 network simulator, with realistic UAV mobility modeled by an extended BOID algorithm.This dataset includes 122,171 labeled flow records among five traffic

Table 1 Summary of literature review: test-time evasion attacks and defenses

Author name	Dataset used	ML/DL Models	Attack type	Attack setting	Evaluation metrices	Baseline findings	After attack findings
Awad et al. [3]	CIC-IDS-2017 CIC-IDS-2018	IDS Classifiers	Carlini and Wagner	White Box	Accuracy, Precision, Recall, F-1Score	Accuracy: 98.11% Precision: 98.11% Recall: 98.11% F1-score: 98.07%	Accuracy: 36% Precision: 30% Recall: 35% F1-score: 32%
Baddukonda et al. [13]	CIFAR, MNIST, ImageNet	LeNet, LR, InceptionV3	Carlini and Wagner & FGSM	White Box	Accuracy	MNIST (LeNet /LR): 98.4% CIFAR-10:80% ImageNet(Inception v3):76–78%	MNIST (FGSM attack):1% CIFAR-10 (C&W attack):0–10% ImageNet(FGSM attack): accuracy reduced by 50% (top-1) and 40% (top-5)
Nazeri et al. [14]	MS-COCO KITTI	Various DETR models	Carlini and Wagner & PGD	White Box	Average Precision (AP)	COCO:0.420,0.433, 0.435,0.449 KITTI: 0.378, 0.352, 0.367, 0.38	COCO:0.070, 0.047, 0.060, 0.034 KITTI:0.034, 0.031, 0.023, 0.032
Juraev et al. [15]	ImageNet-1000, CIFAR-100, and CIFAR-10	ResNet-18/50/152	HopSkip Jump	Black Box	Accuracy	ResNet-18 (ImageNet): 94% ResNet-15 (ImageNet): 94.05%	ResNet-18: fooled with 8.4% noise, ResNet-152:fooled with 28.6% noise
Peng et al. [16]	CIFAR10, CIFAR-100, Image Net	ResNet, WRN	Auto Attack	White Box	Accuracy	CIFAR-10:84.05% ImageNet: 63.87%	CIFAR-10:49.97% ImageNet: 34.96%
Chen et al. [17]	Comma2k19	YOLOV8	Auto PGD	White Box	mAP@50, Precision, Recall, Short-range prediction error	mAP@50 = 95.09%, Precision = 99.00%, Recall = 89.34%	mAP@50 = 95.09%: 93.82% (minor drop)

Table 1 continued

Author name	Dataset used	ML/DL Models	Attack type	Attack setting	Evaluation metrices	Baseline findings	After attack findings
Xu et al. [18]	ImageNet, RESISC-45, PASCAL VOC (2007/2012), ,UCF101 (MARS model)	ResNet50-v2 (ImageNet), Dense Net121 (RESISC-45) Object detection: Faster-RCNN with ResNet-50 Video classification: MARS model	Masked PGD (MPGD) Masked AutoPGD (MAPGD) Masked C&W (MCW) Adaptive BPDA	White Box	Top-1 Accuracy	ImageNet benign: 81.62%, RESISC-45 (DenseNet121) benign: 92.9% PASCAL VOC benign, AP: 49.20%, UCF1 (MARS model)Top-1 accuracy benign: 94.55%	ImageNet:MPGD: 14.35%, MAPGD:9.40%, MCW: 49.57% RESISC-45: MPGD: 3.0%, MAPGD: 1.7% PASCAL VOC: MPGD 6.5% UCF101 (MARS model,Top-1 accuracy): MPGD (5% patch): 8.42%, MPGD (10% patch): 3.96%, MAPGD (5% patch): 18.81%, MAPGD (10% patch):0.00%
Yan et al. [19]	Custom malicious traffic dataset CIC-IDS-2017	IDS Classifiers	PGD,Transfer attacks	White Box Black Box	Accuracy	Local substitute models achieve >0.85 accuracy	Adversarial malicious traffic examples evade IDS detection with up to 80% evade rate
Meyers et al. [20]	Self-Driving Cars dataset, Bone Break Classification Image Dataset	DETR-ResNet50,CNN	FGSM, PGD, BIM	White Box	Attack success rate, Qualitative evaluation	MNIST: 99.96% CIFAR-10: 99.83%	Worst-case adversarial failure rate: MNIST: 10^{-1} failures/sec CIFAR-10: 10^{2} failures/sec Attacks such as DeepFool, HopSkipJump, and Adversarial Patch consistently broke models

Table 1 continued

Author name	Dataset used	ML/DL Models	Attack type	Attack setting	Evaluation metrices	Baseline findings	After attack findings
Merzouk et al. [8]	UNSW-NB15, NSL-KDD, CIC-IoV-2024	DQN,PPO, TRPO,A2C, QRDQN	FGSM,BIM	White Box	Accuracy, FNR	UNSW-NB15:93%-95% NSL-KDD:77%-83% (PPO:0.79–0.80, TRPO: 0.79–0.82) CIC-IoV-2024:100%	UNSW-NB15:A2C: FNR = 0.27 ± 0.17 (FGSM)TRPO:FNR = 0.31 ± 0.24 (FGSM) DQN, PPO, QRDQN: FNR = 0.77–0.90 NSL-KDD: A2C: FNR = 0.54 ± 0.12 (FGSM) TRPO: FNR = 0.67 ± 0.11 (FGSM) DQN, PPO, QRDQN: FNR = 0.77–0.90 CICIoV2024: A2C: Avg FNR ≈ 0.8 at $\epsilon = 0.1$ DQN / QRDQN / PPO: FNR ≈ 0.5–0.6 at $\epsilon = 0.1$ TRPO: Best robustness, avg FNR ≈ 0.37
Muthalagu et al. [4]	UNSW-NB15, CIC-IoMT-2024, CIC-IoT-2023	NN,RF	Carlini & Wagner, Auto-PGD attack	White Box	Accuracy, Attack success rate via ART	RF: 95–97% NN: 93–96%	C&W: 45–60% ASR on NN, 20–35% on RF Auto-PGD: 50–65% on NN, 25–40% on RF

Table 1 continued

Author name	Dataset used	ML/DL Models	Attack type	Attack setting	Evaluation metrices	Baseline findings	After attack findings
Merzouk et al. [5]	NSL-KDD, UNSW-NB15, CIDDS-001	MLP-NN	FGSM, BIM, DeepFool, Carlini & Wagner, JSMA	White Box	TPR	NSL-KDD: 74.20%, UNSW-NB15: 88.01%, CIDDS-001: 99.52%	NSL-KDD: detection rate dropped below 26% for most attacks (C&W L∞ = 37%). UNSW-NB15: detection rate dropped to ∼12% for most attacks. CIDDS-001: detection rate fell below 1% (C&W L∞ = 9.03%, C&W L2 = 35.87%)
Roshan et al. [6]	CIC-IDS-2017	Deep NN Classifier	FGSM,PGD,C& W	White Box	Accuracy, AUC	Accuracy: 98.5% AUC-ROC: 98.47%	FGSM:Accuracy dropped to around 62–65% BIM:Accuracy reduced,around 50–55%. C&W: PS-1: Accuracy 35.5%, AUC-ROC 0.364 PS-2: Accuracy 30.7%, AUC-ROC 0.3169 PS-3:Accuracy 38.6%, AUC-ROC 0.4003

Table 1 continued

Author name	Dataset used	ML/DL Models	Attack type	Attack setting	Evaluation metrices	Baseline findings	After attack findings
Sharma et al. [7]	NSL-KDD	LR, MLP, SVM ,DT, RF, XGBoost, KNN	PGD Transfer, PGD-assisted transfer, ZOO, Boundary, HopSkip Jump	White Box and Black Box	Accuracy	LR: 75.2% MLP: 81.1% SVM-linear: 75.2% SVM-rbf: 76.4% Decision Tree: 81.3% Random Forest: 78.0% XGBoost: 78.0% KNN: 76.9% Label Spreading: 76.9%	LR(PGD $\varepsilon \geq 0.15$): 0%MLP (PGD $\varepsilon \geq 0.18$): 0% SVM-linear (HSJ /Transfer): 10-15% SVM-rbf (HSJ / Transfer): 10-15% Decision Tree (HSJ / Transfer): 10-15% Random Forest (HSJ /Transfer): 10-15% XGBoost (HSJ / Transfer): 10-15% KNN (PGD / HSJ): 30% Label Spreading (PGD / HSJ): 30%
Aloraini et al. [21]	HCRL car-hacking dataset	Baseline DNN ,MTH-IDS, DNN and Decision Tree	FGSM, BIM, PGD, C&W, EAD	White Box	Accuracy	IDS: 95% MTH-IDS: 97%	IDS (FGSM): 44% IDS (BIM): 41% IDS (PGD): 38% MTH-IDS (FGSM): 81% MTH-IDS (BIM): 78% MTH-IDS (PGD): 72%
Gungor et al. [22]	MQTT, WUSTL-IIoT, X-IIoTID, Edge-IIoT	NB, DT, RF, XGB, DNN, CNN, RNN	FGSM, BIM, PGD, Deep-Fool, JSMA, EN, ZOO, Transfer	White Box Black Box	Macro F1 score	Baseline: 99%	FGSM: 2% BIM: 4% PGD: 4% DeepFool: 2% JSMA: 65% EN: 57% ZOO: 65%

Table 1 continued

Author name	Dataset used	ML/DL Models	Attack type	Attack setting	Evaluation metrices	Baseline findings	After attack findings
Dinh et al. [9]	CIC-DDoS-2019,IoTID20	SAINT,CDNN, TabNet	FGSM,BIM, DeepFool,Carlini & Wagner	White Box	Accuracy	TabNet, CIC-DDoS-2019: 95.03% TabNet, IoTID20: 93.47%	TabNet, CIC-DDoS2019: FGSM: 94.72%, BIM: 94.11%, DeepFool: 84.20%, CW: 26.12% TabNet, IoTID20: FGSM:93.21%, BIM:92.13%, DeepFool: 84.96%, CW: 53.72%
Meyers et al. [10]	Synthetic numeric datasets (Gaussian), KDD-NSL, Truthseeker Twitter dataset	Kernelized SVMs (Linear, Polynomial, RBF)	Massively parallel Projected Gradient Descent (PGD)	White Box	Benign accuracy, Adversarial error, False high-confidence misclassifications, Attack Time	Baseline: 100%	PGD attack: adversarial error ∼6× higher than benign error Adversarial retraining: fails to restore baseline (still far below 100%)
Aloraini et al. [23]	Car Hacking Dataset	DNN, DT,RF,ET, XGBoost	FGSM,BIM, PGD, DT	White Box	Baseline Accuracy, Attack Success Rate (ASR)	DNN: 100% Decision Tree: 100% Random Forest: 100% Extra Trees: 100% XGBoost: 100%	Gradient-based attack (FGSM/BIM/PGD): DNN: 99% Decision Tree: 11% Random Forest: 34% Extra Trees: 25% XGBoost: 51%
Kumar et al. [11]	CIC-IDS-2017	DNN	FGSM,BIM, PGD,DeepFool, JSMA	White Box	DRR, ACC, FP, Error Rate	Baseline: 99%	FGSM:100% detected BIM:99.85% detected PGD: 99.91% detected DeepFool: 99.99% detected JSMA: 100% detected

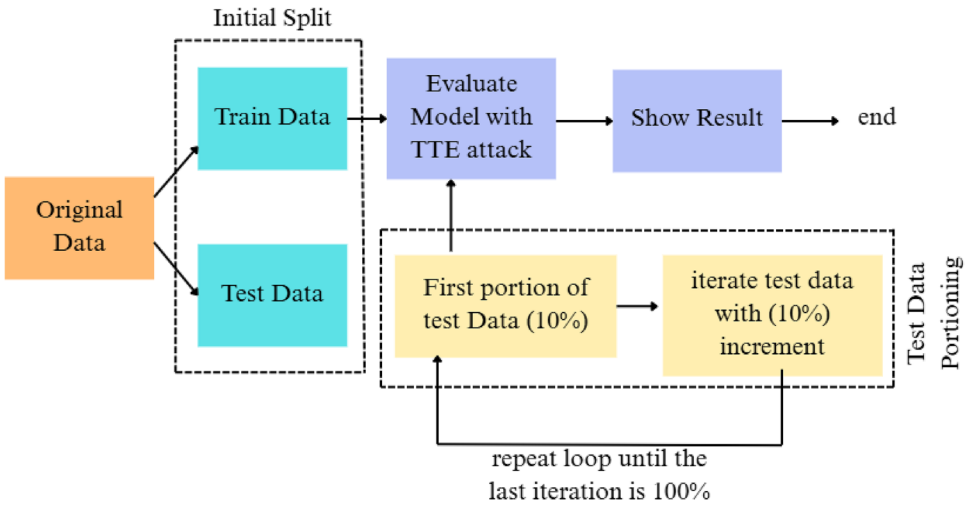


Fig. 1 Workflow diagram for scalability checking

classifications such as Normal, Blackhole, Flooding, Sybil, and Wormhole attacks. data samples represents 22 features grouped into connection, traffic volume and performance metrics. Also it enables the evaluation of machine learning-based IDS under various scenarios, including imbalanced attack distributions and swarm mobility making it the best newest option for adversarial robustness experiments like us.

Our dataset selection and the distribution strategy applied to the testing data ensure a broad and diverse scope for the study, encompassing both traditional and IoT-specific environments. This approach maintains computational feasibility while preserving sufficient diversity in attack representation.

3.2 Data preprocessing

All three datasets, CICIoMT-2024, UNR-IDD, and UAV-IDS-2025, were preprocessed using a MinMaxScaler to normalize feature values to a common range. Since the datasets contain a variety of categorical attributes, we applied LabelEncoder to transform these categorical labels into numerical form suitable for model training. No class balancing techniques were applied to the datasets. This was a deliberate choice, as our objective was to evaluate model performance under naturally occurring class distributions, which better reflects real-world deployment conditions where imbalance is inherent to network intrusion and anomaly data.

3.3 Train-test split

The datasets used in this study were provided in pre-partitioned train and test subsets by their original sources, preserving the intended evaluation protocol for each. Rather than imposing a uniform split ratio across all datasets, the original partitioning was retained, allowing model performance to be assessed under diverse and realistic evaluation conditions specific to each dataset. For instance, UNR-IDD follows an approximate 82:18 train-test ratio, UAV-IDS follows an approximate 70:30 ratio, and CICIoMT-2024 employs a comparatively smaller, fixed-size test set per attack category evaluated against a much larger training pool. This variation across datasets reflects the diversity of real-world data availability scenarios and strengthens the generalizability of our evaluation, as the model is tested under differing train-test proportions rather than a single, potentially favorable configuration. Table 2, provides the data distribution of the datasets.

3.4 Evaluation metrics

We have selected Accuracy as our primary evaluation metrics to assess the performance of the intrusion detection models.The general definition of accuracy for a classifier is expressed as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100\%, \tag{1}$$

where TP and TN represent the number of true positives and true negatives, while FP and FN denote false positives and false negatives. To get a straight forward measure, accuracy allows comparison between clean and adversarial scenarios.

3.5 Classifiers

Researchers have predominantly relied on regression-based and tree-based classifiers when evaluating evasion attacks, as these models are widely used in traditional machine learning tasks, particularly for tabular data, and provide higher interpretability compared to deep neural networks. For instance, Sharma et al. [7] and Kumar et al. [11] adopted such classifiers in their adversarial evaluations.

To broaden the applicability of our analysis, we selected both conventional and contemporary classifiers. A **PyTorch-**

Table 2 Data distribution across datasets

Dataset	Subset	Train data	Test data
UNR-IDD		Benign 254 PortScan 666 TCP-SYN 651 Blackhole 561 Diversion 414 Overflow 73	Benign 49 TCP-SYN 156 PortScan 127 Blackhole 122 Diversion 90 Overflow 17
UAV-IDS		Benign 18320 Blackhole Attack 18277 Wormhole Attack 18260 Sybil Attack 16854 Flooding Attack 13808	Benign 7852 Blackhole Attack 7833 Wormhole Attack 7826 Sybil Attack 7223 Flooding Attack 5918
CICIOMT	Recon-MQTT	Benign 3846 Recon-OS_Scan 3846 TCP_IP-DDoS-UDP1 3846 TCP_IP-DDoS-UDP2 3846 MQTT-DDoS-Publish_Flood 3846 MQTT-DoS-Publish_Flood 3846 MQTT-DDoS-Connect_Flood 3846 MQTT-Malformed_Data 3846 TCP_IP-DDoS-ICMP1 3846 MQTT-DoS-Connect_Flood 3846 ARP_Spoofing 3846 Recon-VulScan 2173 Recon-Ping_Sweep 740	Benign 269 Recon-OS_Scan 269 TCP_IP-DDoS-UDP1 269 TCP_IP-DDoS-UDP2 269 Recon-VulScan 269 MQTT-DDoS-Publish_Flood 269 MQTT-DDoS-Connect_Flood 269 MQTT-Malformed_Data 269 TCP_IP-DDoS-ICMP1 269 MQTT-DoS-Publish_Flood 269 ARP_Spoofing 269 MQTT-DoS-Connect_Flood 269 Recon-Ping_Sweep 186
	DOS-Benign	Benign 192732 TCP_IP-DoS-UDP 566950 TCP_IP-DoS-SYN 441903 TCP_IP-DoS-ICMP 416292 TCP_IP-DoS-TCP 380384	Benign 600 TCP_IP-DoS-ICMP 600 TCP_IP-DoS-SYN 600 TCP_IP-DoS-TCP 600 TCP_IP-DoS-UDP 600
	DDOS-Benign	Benign 192732 TCP_IP-DDoS-UDP 1635956 TCP_IP-DDoS-ICMP 1537476 TCP_IP-DDoS-TCP 804465 TCP_IP-DDoS-SYN 801962	Benign 700 TCP_IP-DDoS-ICMP 700 TCP_IP-DDoS-SYN 700 TCP_IP-DDoS-TCP 700 TCP_IP-DDoS-UDP 700
	ARP-Benign	Benign 192732 ARP 16047	ARP 1700 Benign 1700

Table 3 Model architecture (PyTorch Classifier)

Layer	Type	Output size
Input layer	Fully connected (Linear)	64
Activation	ReLU	64
Output layer	Fully connected (Linear)	$n_classes$

Table 4 Training hyperparameters (PyTorch Classifier)

Hyperparameter	Value
Optimizer	Adam
Learning Rate	0.001
Loss Function	Cross-Entropy Loss
Number of Epochs	1200
Batch Size	64

based fully connected neural network was chosen as the primary classifier to align with the architecture supported by the ART library proposed by Nicolae et al. [2]. Employing this classifier allows us to assess the intrinsic performance of ART within a tabular data setup. In addition, we incorporated a **1D Convolutional Neural Network (CNN)** to extend the applicability of the ART framework across a different model paradigm, one capable of capturing local feature interactions through convolutional operations rather than relying solely on fully connected transformations. Both classifiers were trained using the Adam optimizer, as it combines the benefits of adaptive learning rates and momentum, enabling faster and more stable convergence compared to standard stochastic gradient descent.

3.5.1 Model architecture and training configuration of PyTorch classifier

The primary classifier was implemented as a fully connected feedforward neural network using PyTorch. The network consists of a single hidden layer with ReLU activation, followed by an output layer that produces class scores corresponding to the number of target classes. The complete layer structure is presented in Table 3, while the corresponding training hyperparameters are summarized in Table 4. The model was trained using the Adam optimizer with the Cross-Entropy loss function over 1200 epochs, using a batch size of 64 and mini-batch shuffling at every epoch to promote stable convergence.

3.5.2 Model architecture and training configuration of CNN classifier

The second classifier is a 1D Convolutional Neural Network (CNN) designed to process tabular network traffic features by treating each feature vector as a single-channel 1D signal. The network comprises two convolutional blocks, each followed by a ReLU activation, with a max-pooling operation applied after the first block to reduce the spatial dimensionality of the extracted features. The resulting feature maps are flattened and passed through two fully connected layers to produce the final class predictions. Table 5 presents the complete layer-wise configuration, including the input/output dimensions and layer-specific parameters at each stage. Similar to the PyTorch classifier, this model was trained using the Adam optimizer with a learning rate of 0.001 and the Cross-Entropy loss function, over 50 epochs with a batch size of 64.

3.6 Performed attacks

We applied the ART to perform TTE attacks. The attacks were conducted under both white-box and black-box scenarios. In the white-box setting, we employed AutoAttack and Auto-PGD proposed by Croce et al. [24], as well as the C&W attack introduced by Carlini et al. [25]. For the black-box scenario, the HSJ attack proposed by Chen et al. [26] was selected. These methods were chosen to encompass both gradient-based and decision-based attack strategies, thereby ensuring a comprehensive evaluation of evasion robustness.

Table 5 CNN model architecture

Layer	Type / Activation	Parameters
Conv1	1D Convolution	1→16 ch, kernel=3, pad=1
Activation 1	ReLU	–
MaxPool	1D Max Pooling	kernel=2, stride=2
Conv2	1D Convolution	16→32 ch, kernel=3, pad=1
Activation 2	ReLU	–
Flatten	Flatten Layer	–
FC1	Fully Connected (Linear)	$32\times n_f/2 \rightarrow 128$
Activation 3	ReLU	–
Output Layer	Fully Connected (Linear)	$128 \rightarrow n_classes$

Table 6 Results found on UNR-IDD database

Classifier used	Attack used	Clean accuracy	Adversarial accuracy
Pytorch	Auto PGD Attack	80.21%	19.07%
	Auto Attack	80.21%	26.56%
	C& W Attack	82%	13.73%
	HSJ Attack	80.39%	13.19%
CNN	Auto PGD Attack	73.26%	34.94%
	Auto Attack	73.26%	40.29%
	C& W Attack	73.26%	27.27%
	HSJ Attack	73.26%	16.58%

Hyperparameter tuning plays a critical role across different attack scenarios and datasets. Complex datasets may require stronger perturbations for successful evasion; however, excessive perturbation can lead to model instability or collapse. Furthermore, optimal hyperparameter settings often vary depending on the dataset characteristics. From an experimental standpoint, however, adjusting hyperparameters for each dataset may obscure the intrinsic behavior of the data under adversarial conditions. Therefore, to ensure fair comparison and consistency across datasets, the attack hyperparameters were kept fixed throughout the experiments.

Tables 7, 8, 9 and 10 summarize the hyperparameters used in our experiments with Auto Attack, Auto PGD Attack, Carlini and Wagner Attack, and HopSkipJump Attack.

Table 7 Hyperparameters used for auto attack

Attack type	Hyperparameter
Auto Attack	estimator=classifier, norm=np.inf, eps=0.03, eps_step=0.001, batch_size=32, attacks=attacks_tabular

4 Results

4.1 Attack performance

We have observed a significant difference between clean performance and performance under attack. Across all datasets, the classifiers demonstrated a robust baseline accuracy ranging from 70 to 100%, depending on the dataset. However, these results varied in different attack scenarios.

For the UNR-IDD dataset, models trained on the clean dataset showed baseline accuracies of approximately 73.26% with the CNN classifier and 81% with the PyTorch classifier. After conducting TTE, the results differed based on the attack situation. Notably, the PyTorch-based model achieved a baseline accuracy of 82% on clean data but dropped to 13.73% under the C&W attack. Similarly, the HSJ attack reduced accuracy to 13.19%. In contrast, CNN models demonstrated relative resilience in specific scenarios, particularly under AutoAttack, where the accuracy was maintained at 40.29%. This observation indicates that the UNR-IDD dataset has a moderate level of accuracy, which reflects an extreme class imbalance across the labels. This imbalance results in moderated accuracy across the classifiers because the training was insufficient due to a limited number of data instances.

Overall, the results reveal that the PyTorch model collapses under adversarial stress due to overfitting and sensitivity to gradients, despite initially achieving higher accuracy on clean data. Conversely, the CNN model exhibits partial robustness because its convolutional features are stronger against perturbations. Additionally, the numeric nature of the UNR-IDD dataset enhances its vulnerability to adversarial attacks. This issue could be addressed by using synthetic data to balance the label disparities and to scale the features effectively. Table 10 illustrates the results achieved during the TTE with the UNR-IDD dataset.

For the CICIoMT-2024 dataset, we have four subsets: Benign-ARP, Benign-DOS, Benign-DDoS, and Benign-MQTT. In the Benign-ARP subset, models trained on clean data achieved 82.68% accuracy with the PyTorch classifier and 76.53% with the CNN classifier. This dataset primarily consists of benign ARP traffic patterns with limited attack diversity, resulting in moderate accuracy due to limited feature variance and a smaller representation of complex samples during training.

After applying TTE, the results varied across different attack methods. The PyTorch model experienced a significant drop in performance under the Auto-PGD and C&W attacks, with accuracies of only 17% and 19.12%, respectively. In contrast, AutoAttack maintained an accuracy of 24.56%, while HSJ reached 49.97%. The CNN model demonstrated moderate robustness, with accuracies ranging from 27.85 to 49.71% across the various attacks.

Overall, these results indicate that while the PyTorch model initially attained a higher clean accuracy, it demon-

Table 8 Hyperparameters used for Auto PGD Attack

Attack type	Hyperparameter
Auto PGD	estimator=classifier, norm=np.inf, eps=0.05, eps_step=0.005, max_iter=10, targeted=False, batch_size=32

Table 9 Hyperparameters used for Carlini and Wagner Attack

Attack type	Hyperparameter
C & W	classifier=classifier, norm=L2 norm confidence=0.15, targeted=False, learning_rate=0.01, binary_search_steps=3, max_iter=150, initial_const=0.02

Table 10 Hyperparameters used for HopSkipJump Attack

Attack type	Hyperparameter
HSJ	max_iter = 10 max_eval = 100 init_eval = 1 init_size = 10

strated significant vulnerability under adversarial conditions due to overfitting and sensitivity to gradients. In contrast, the CNN model exhibited better stability, as convolutional feature extraction provides stronger local feature invariance. To address the limitations in dataset structure and imbalance, it is recommended to expand the training samples and apply appropriate feature scaling to enhance robustness against adversarial perturbations.

For the DOS Benign subset, both the PyTorch and CNN models achieved nearly perfect clean accuracies, with 99.93% for PyTorch and 100% for CNN. However, after applying TTE, the PyTorch model exhibited strong resilience against Auto-PGD and AutoAttack, with only a slight accuracy drop to above 94%. However, under the C & W and HSJ attacks, the accuracy dropped to 21.07% and 0.13%, respectively. Similarly, the CNN model maintained high accuracy under AutoAttack and Auto-PGD but saw a decrease with C & W (14.53%) and HSJ (34.79%). This suggests that both models effectively learned the characteristics of DOS Benign traffic, but they are comparatively weak against optimization-based and boundary-based attacks like Hop-Skip Jump and C & W. This indicates that the Benign-DOS subset exhibits repetitive feature structures and low feature diversity, which makes the models vulnerable to precise adversarial perturbations.

In the case of the DDOS Benign subset, both the PyTorch and CNN classifiers performed well on clean data, achieving 100% accuracy. However, adversarial attacks revealed substantial weaknesses in feature generalization. The PyTorch model maintained strong performance under Auto-PGD and AutoAttack, achieving over 67% accuracy, but its performance collapsed under HSJ (4.29%) and C & W (22.51%), indicating issues with gradient sensitivity and weak decision boundary smoothness. The CNN classifier displayed a similar pattern, preserving 84.91% accuracy under HSJ but dropping to 26.63% under C & W. These results, which are also seen in the DOS Benign subset, show that even though the patterns were well learned, their limited variation allows for adversarial manipulation, exploiting breaches in the latent feature space.

Table 11 illustrates the results achieved during the TTE in the CICIOMT-2024 dataset.

In the evaluation of the UAV-IDS 2025 dataset, the PyTorch classifier achieved clean accuracies ranging from 94 to 96.05%, while the CNN classifier achieved a clean accuracy of 99.08%. This suggests that the features are well-distributed, and the numerical attributes are strongly correlated, which facilitates effective training for the models. However, when tested against Test Time Evasion (TTE) attacks, the performance of the PyTorch classifier significantly declined. Specifically, Auto-PGD reduced its accuracy to 6.40%, AutoAttack to 16.23%, C&W to 5.31%, and HSJ to 10%. In contrast, the CNN classifier exhibited stronger resilience, achieving accuracies of 14.37% with Auto-PGD, 55% with AutoAttack, 0.99% with C&W, and 19.96% with HSJ. Although CNNs demonstrated some resistance to various attacks, the optimization-focused approach of C&W proved to be particularly damaging, highlighting the dataset's vulnerability to targeted adversarial attacks.

Table 12 illustrates the results obtained during the TTE in the UAV-IDS dataset. These findings emphasize that, despite achieving high clean accuracy and effective feature learning during training, adversarial perturbations can severely undermine model performance.

A comprehensive scalability analysis was performed under **PyTorch classifiers**. This experiment systematically increased the proportion of testing data from 10 to 100% with 10% increments per attack, enabling the observation of adversarial performance under progressive data representations. The same TTE attacks as in the Attack Performance evaluation were used: Auto-PGD, AutoAttack, C&W, and HSJ Attack.

For the **UNR-IDD dataset**, the clean accuracy of PyTorch classifiers remained consistently above 78%. However, adversarial robustness decreased substantially as the dataset scale increased. Under Auto-PGD, adversarial accuracy

Table 11 Results found on CICIOMT-2024 database

Section	Classifier Used	Attack Used	Clean Accuracy	Adversarial Accuray	Section	Classifier Used	Attack Used	Clean Accuracy	Adversarial Accuray
Benign ARP	Pytorch	Auto PGD Attack	82.68%	17%	DDOS Benign	Pytorch	Auto PGD Attack	100%	67.77%
		Auto Attack	83.56%	24.56%			Auto Attack	100%	99.77%
		C & W	81.94%	19.12%			C& W Attack	100%	22.51%
		Hop Skip	80.63%	49.97%			HSJ Attack	100%	4.29%
	CNN	Auto PGD Attack	76.53%	31.82%		CNN	Auto PGD Attack	100%	56.29%
		C & W	76.53%	27.85%			C& W Attack	100%	26.63%
		HSJ Attack	76.53%	49.71%			Auto Attack	100%	19.91%
		Auto attack	76.53%	44.21%			HSJ Attack	100%	84.91%
DOS Benign	Pytorch	Auto PGD Attack	99.93%	99.77%	MQTT Recon	Pytorch	Auto PGD Attack	70.62%	36.35%
		Auto Attack	100%	94.86%			Auto Attack	70.27%	11.95%
		C & W	99.93%	21.07%			C& W Attack	70.24%	35%
		HSJ Attack	100%	0.13%			HSJ Attack	70.18%	11.63%
	CNN	Auto PGD Attack	100%	34.37%		CNN	Auto PGD Attack	70.80%	33.27%
		C & W	100%	14.53%			C& W Attack	70.80%	38.72%
		Auto Attack	100%	95.34%			Auto Attack	70.80%	54.39%
		HSJ Attack	100%	20%			HSJ Attack	70.80%	13.56%

Table 12 Results found on UAV-IDS database

Classifier used	Attack used	Clean accuracy	Adversarial accuracy
Pytorch	Auto PGD Attack	94.90%	6.40%
	Auto Attack	95.81%	16.23%
	C& W Attack	94.00%	5.31%
	HSJ Attack	96.05%	10.00%
CNN	Auto PGD Attack	99.08%	14.37%
	C& W Attack	99.08%	0.99%
	HSJ Attack	99.08%	19.96%
	Auto Attack	99.08%	55.52%

dropped from 74.15% (10% testing data) to 37.97% (100% testing data), while AutoAttack showed a similar decline from 73.44 to 28.52%. The C&W attack proved even more severe, reducing accuracy from 72.55% to 17.83%. HSJ initially showed 73.98% at 10% testing data, but converged to only 22.99% at full scale (100%). Figure 2a shows the trend achieved by the increasing testing data of UNR-IDD dataset.

In **CICIoMT-2024 (Benign ARP)**, baseline accuracy revolved around 79–80%. Auto-PGD showed a gradual decline from 75.53% (10% testing data) to 48.24% (100% testing data), while AutoAttack was more destructive, reducing accuracy from 75.91% to 35.59%. The C&W attack degraded performance from 74.76% to 26.88%. Interestingly, HSJ caused a slower reduction, with adversarial accuracy maintained at 49.88% with full dataset. Figure 2b shows the trend achieved by the increasing testing data of Benign-ARP sunset of CICIOMT-2024 dataset.

The **DoS and DDoS benign subsets** revealed sharp contrasts. For DoS, clean accuracy was 100% across all scales. AutoAttack preserved near-perfect accuracy above 98% in all partitions, whereas both C&W (92.23% to 21.43%) and HSJ (91.77% to 19.67%) reduced performance drastically. The DDoS subset exhibited similar trends, with AutoAttack maintaining high accuracy from 98.86% to 89.03%, while C&W dropped from 92.09% (10%) to 19.71% (100%), and HSJ ranged from 92.03% to 20.20%. Figure 2c and 2d shows the trend achieved by the increasing testing data of Benign-DOS and Benign-DDOS subset of CICIOMT-2024 dataset respectively.

The **MQTT Recon subset** proved particularly vulnerable. Clean accuracy started at 70%. Auto-PGD dropped from 66.14% (10% testing data) to 35.00% (100% testing data), while AutoAttack caused a steep decline from 73.44% to 28.52%. The C&W attack followed a similar trend (67.19% to 34.48%), and HSJ was the most destructive, reducing adversarial accuracy from 64.24% at 10% to just 8.82% at full scale. Figure 2e shows the trend achieved by the increasing testing data of Benign-DOS and Benign-MQTT and Recon subset of CICIOMT-2024 dataset.

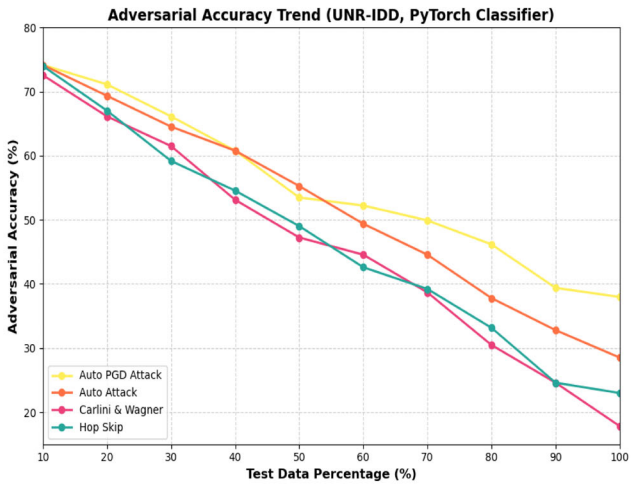
For the **UAV-IDS 2025 dataset**, the clean accuracy of PyTorch classifiers remained very high 96.05%, indicating excellent baseline performance and well-learned feature correlations. However, adversarial robustness decreased sharply as the test dataset scale increased. Under Auto-PGD, accuracy dropped progressively from 86.15% at the smallest subset to 6.51% at the full scale. AutoAttack followed a similar trend, decreasing from 87.83% to 16.27%. The C&W attack portrayed a more severe effect, reducing accuracy from 86.27% to 5.31%, while HSJ initially started at 86.60% and declined to 1.91% at full scale. Figure 2f shows the trend achieved by the increasing testing data of the UAV-IDS 2025 dataset.

From the figures we can observe the downward trend of the performance accuracy of the databases with the testing data being increased. This illustrates that attacks like Auto PGD and Auto Attack maintains a less steeper slope while increasing the testing data on the other hand, attacks like C&W and HSJ are particularly destructive and follows a steeper slope at larger scales, which highlights the need for scalable defense strategies and robustness improvements against these attacks.

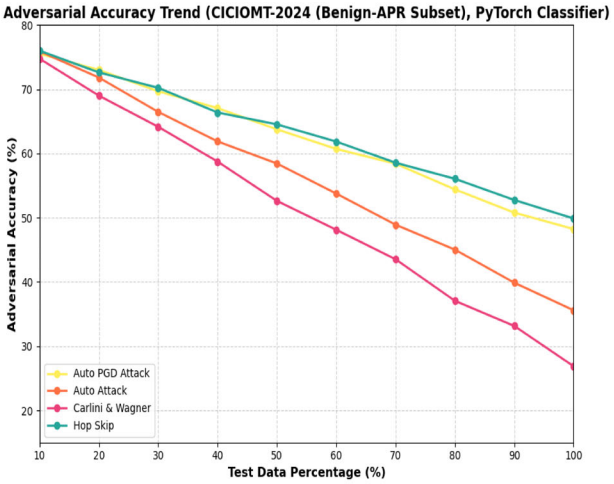
5 Limitations and future works

We conducted experiments using four types of attacks: AutoAttack, Auto-PGD, C&W, and HSJ. During implementation on larger datasets, several computational challenges were encountered. In particular, the C&W and HSJ attacks exhibited significant time complexity. These attacks required substantial execution time, especially when applied to larger subsets of the CICIMOT-2024 dataset. In some cases, the models became unstable or failed to converge due to the combination of large testing data volumes and prolonged execution time.

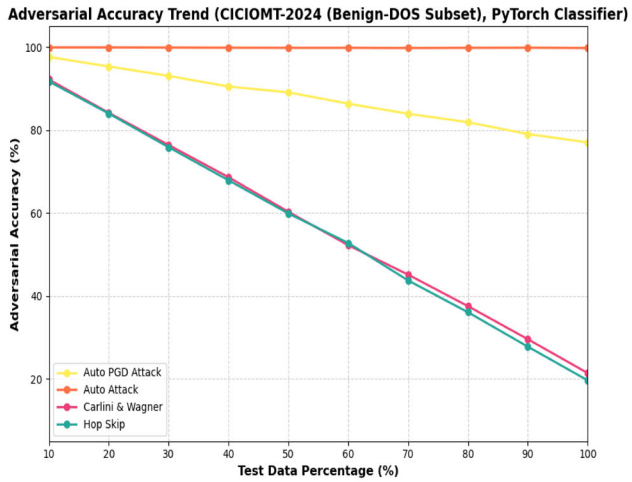
To address this issue, we partitioned the testing data into smaller subsets and conducted experiments iteratively. Although this approach mitigated the model collapse issue, the time complexity remained considerable. This behavior can be attributed to the iterative, optimization-based nature



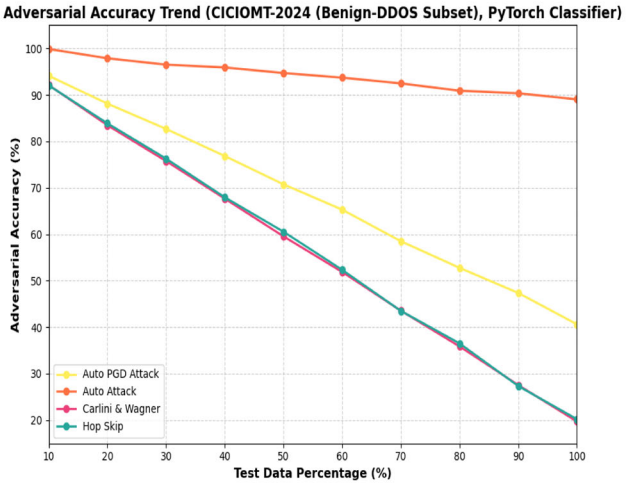
Accuracy Trend for UNR-IDD Dataset



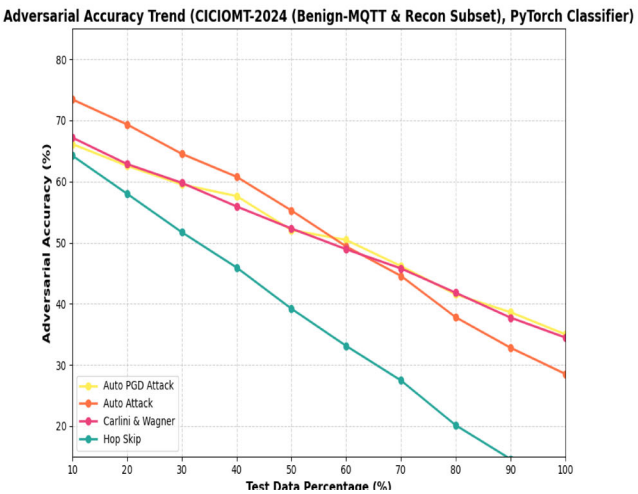
Accuracy Trend for Benign-ARP (CICIOMT-2024)



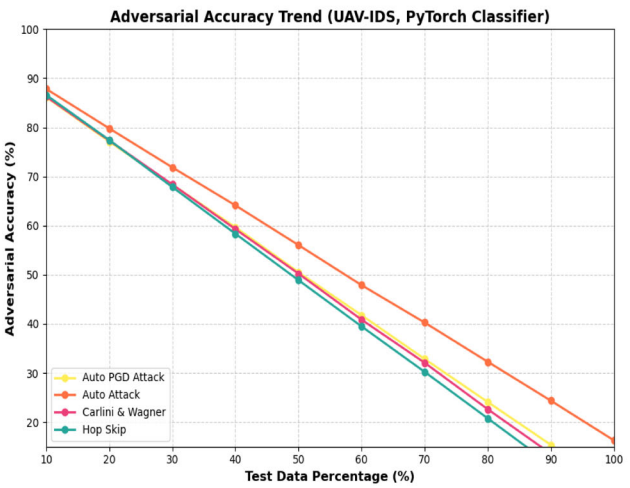
Accuracy Trend for Benign-DoS (CICIOMT-2024)



Accuracy Trend for Benign-DDoS (CICIOMT-2024)



Accuracy Trend for Benign-MQTT and Recon (CICIOMT-2024)



Accuracy Trend for UAV-IDS 2025

Fig. 2 Comparative accuracy trends across different dataset subsets

of these attacks, which aim to identify the minimal adversarial perturbation required to misclassify. Such optimization processes inherently require extensive computation. Despite the computational overhead, the C&W attack demonstrated strong effectiveness in white-box scenarios, achieving high perturbation success rates even with minimal hyperparameter tuning.

Another implementation challenge was observed with the UAV-IDS-2025 dataset when using the CNN classifier. The input data type had to be converted from "float64" to "float32" to ensure compatibility with the deep learning framework. While this conversion reduced numerical precision, it introduced minor rounding effects that may influence optimization-intensive attacks, particularly C&W attack.

These limitations highlight important practical considerations in adversarial experimentation. Future work will focus on reducing computational overhead in optimization-based attacks and investigating strategies to mitigate the impact of numerical precision conversion on adversarial robustness evaluation.

6 Conclusion

This study extended contemporary TTE evaluation methodologies to three modern intrusion detection datasets, UNR-IDD, CICIOMT-2024, and UAV-IDS, to assess the practical applicability of the ART within NIDS. A CNN classifier and a PyTorch classifier were evaluated under a diverse set of adversarial threat models, including AutoAttack, Auto-PGD, and C&W in white-box settings, and HSJ in black-box scenarios. The empirical results reveal a significant susceptibility to adversarial manipulation, with the C&W and HSJ attacks causing up to an 80% reduction in classification accuracy, even for models exhibiting strong baseline performance.

A scalability analysis further demonstrated that, under adversarial conditions, model accuracy consistently declines as the test set size increases, contrasting the conventional expectation that larger datasets primarily benefit model generalization. These observations underscore two central insights. First, adversarial robustness in intrusion detection is inherently context-dependent, shaped not only by model architecture but also by dataset structure, feature distribution, and underlying task characteristics. Second, dataset size alone is insufficient to ensure robustness. Instead, resilience arises from the combined influence of dataset design, model properties, and the specific adversarial threat model. Overall, the findings highlight the necessity for adaptive, context-aware learning and defense mechanisms that can sustain dependable performance under sophisticated test-time adversarial conditions. Future work should investigate integrated robustness frameworks that jointly consider data properties, model architecture, and dynamic adversarial behavior to enhance the reliability and security of next-generation NIDS.

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Data Availability No datasets were generated or analysed during the current study.

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